PAIRR attitudes paper analysis

Sophia Minnillo

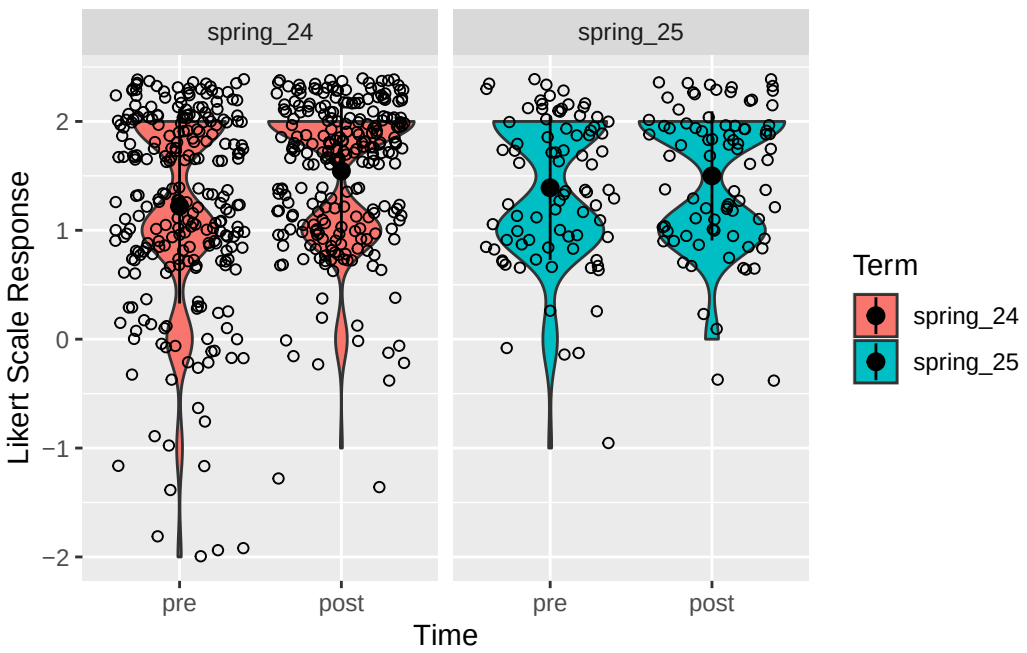
2026-03-25

Set up

Descriptive Statistics

1. I understand the limitations and advantages of AI writing tools like ChatGPT

Graph



Table

```
Automatically converting the following non-factors to factors: Term_pre
```

Term_pre	time	N	value	sd	se	ci	plus_sigma	minus_sigma
spring_24	pre	244	1.236888	1.2637692	0.0809045	0.1593637	2.500657	-0.0268814
spring_24	post	244	1.560658	0.8912196	0.0570545	0.1123845	2.451878	0.6694386
spring_25	pre	82	1.343895	0.9363997	0.1034080	0.2057496	2.280295	0.4074953
spring_25	post	82	1.453651	0.8388705	0.0926377	0.1843201	2.292521	0.6147806

Table #2: Descriptive Stats in Paper

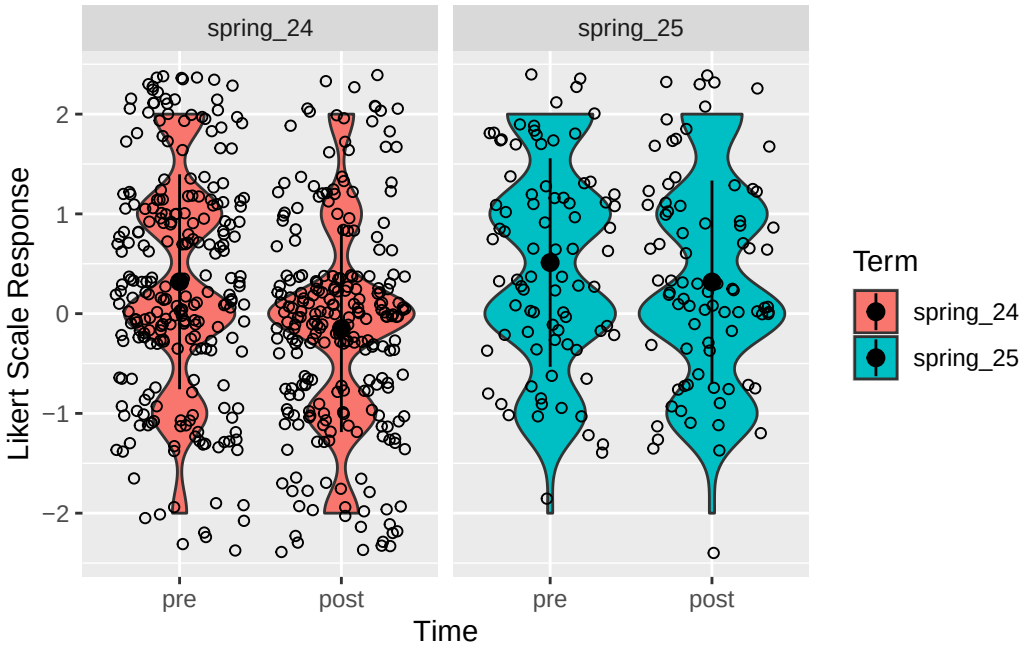
```
`summarise()` has regrouped the output.
i Summaries were computed grouped by time and Term_pre.
i Output is grouped by time.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(time, Term_pre))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

time	Term_pre	median	mean	sd	count
pre	spring_24	1	1.221312	0.8936198	244
pre	spring_25	1	1.390244	0.6621345	82
post	spring_24	2	1.545082	0.6301874	244
post	spring_25	2	1.500000	0.5931710	82

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by time, Term_pre, and VALUE.
i Output is grouped by time and Term_pre.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(time, Term_pre, VALUE))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

2. Using an AI Writing Tool on a Writing Assignment is a form of Cheating/Plagiarism

Graph



Table

```
Automatically converting the following non-factors to factors: Term_pre
```

Term_pre	time	N	value	sd	se	ci	plus_sigma	minus_sigma
spring_24	pre	241	0.4031705	1.523150	0.0981147	0.1932760	1.926321	-1.119980
spring_24	post	241	-0.0657092	1.468055	0.0945658	0.1862849	1.402346	-1.533765
spring_25	pre	82	0.2662916	1.478135	0.1632327	0.3247820	1.744427	-1.211844
spring_25	post	82	0.0711697	1.438072	0.1588085	0.3159791	1.509242	-1.366903

Table #2: Descriptive Stats in Paper

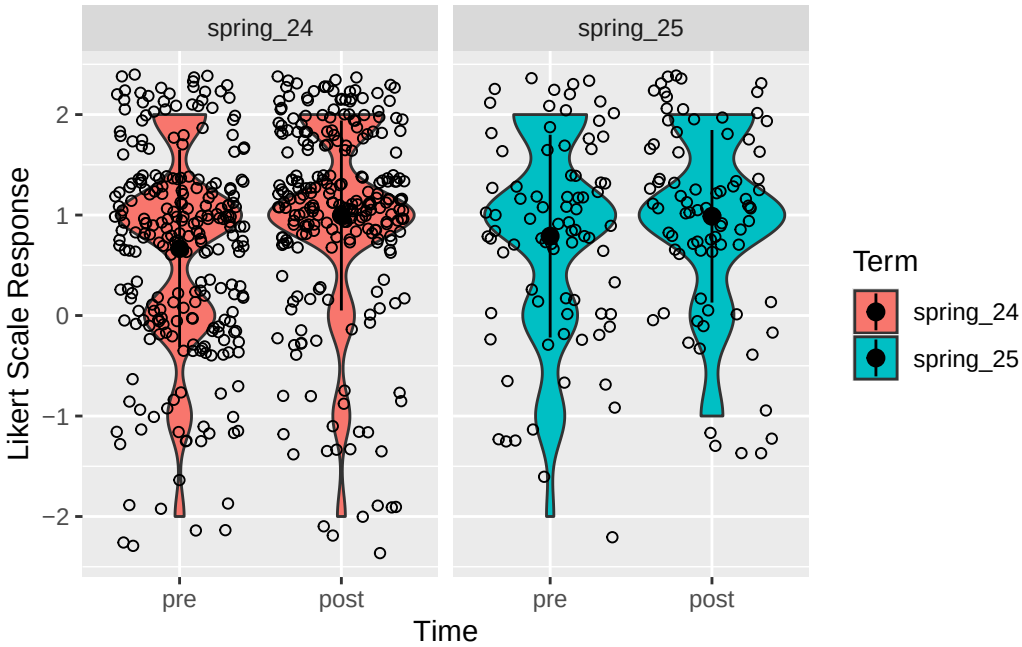
```
`summarise()` has regrouped the output.
i Summaries were computed grouped by time and Term_pre.
i Output is grouped by time.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(time, Term_pre))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

time	Term_pre	median	mean	sd	count
pre	spring_24	0.0	0.3195021	1.077030	241
pre	spring_25	0.5	0.5121951	1.045200	82
post	spring_24	0.0	-0.1493776	1.038072	241
post	spring_25	0.0	0.3170732	1.016871	82

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by time, Term_pre, and VALUE.
i Output is grouped by time and Term_pre.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(time, Term_pre, VALUE))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

3. Peer-feedback on my writing is clear and/or helpful

Graph



Table

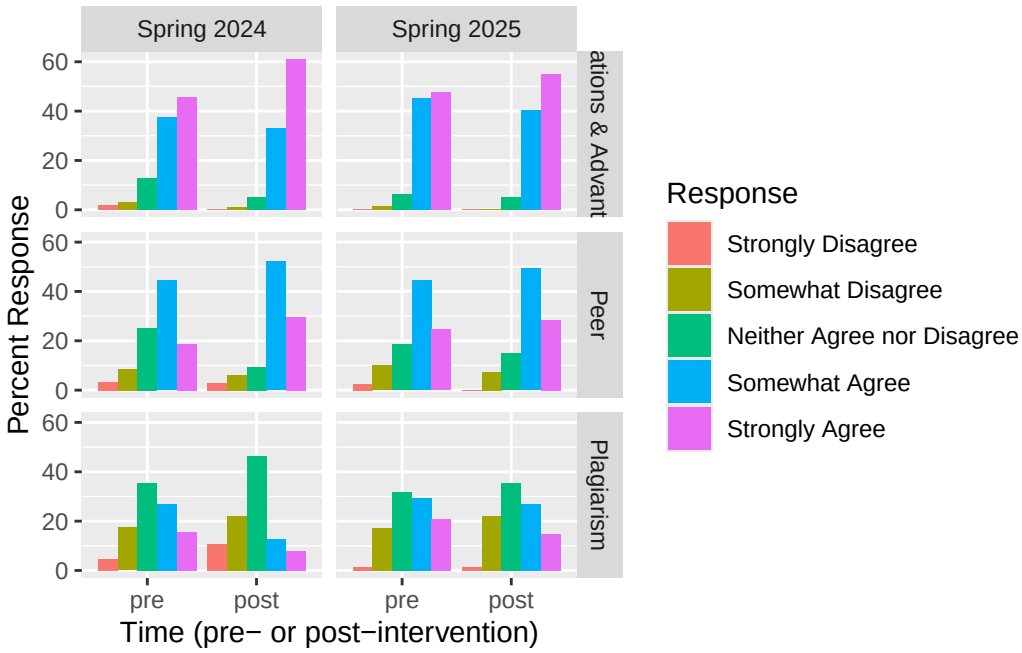
```
Automatically converting the following non-factors to factors: Term_pre
```

Term_pre	time	N	value	sd	se	ci	plus_sigma	minus_sigma
spring_24	pre	246	0.6809316	1.389954	0.0886202	0.1745548	2.070886	-0.7090226
spring_24	post	246	1.0101999	1.337057	0.0852477	0.1679118	2.347257	-0.3268573
spring_25	pre	81	0.7468003	1.426816	0.1585351	0.3154948	2.173616	-0.6800152
spring_25	post	81	0.9443312	1.214369	0.1349299	0.2685190	2.158700	-0.2700375

Table #2: Descriptive Stats in Paper

```
`summarise()` has regrouped the output.
`summarise()` has regrouped the output.
i Summaries were computed grouped by time and Term_pre.
i Output is grouped by time.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(time, Term_pre))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Graph for all 3



LMER Models

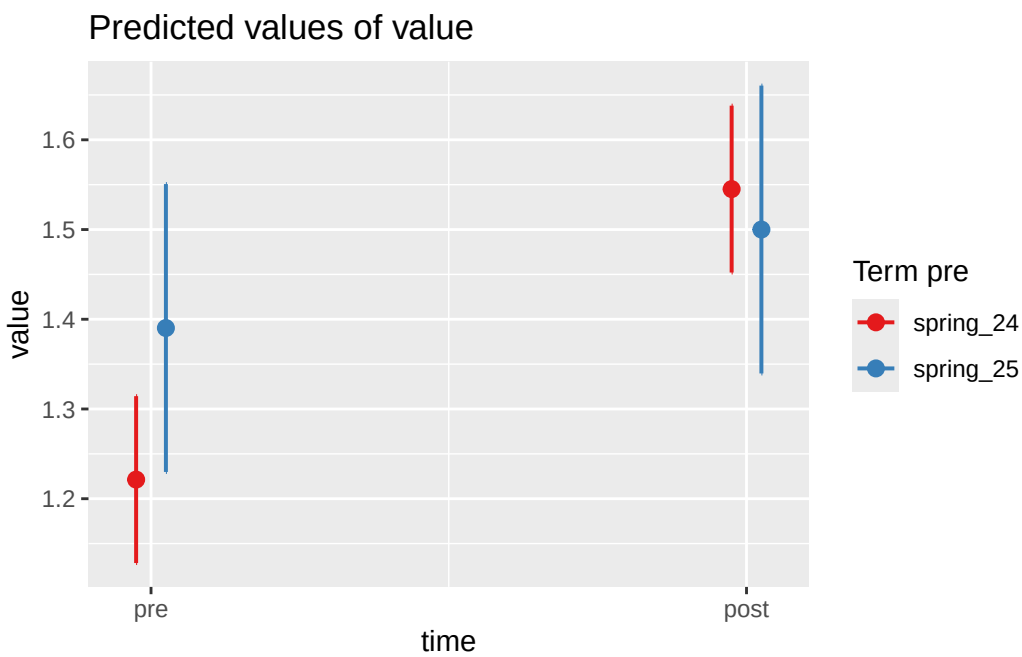
We ended up not going with these because ordinal data violates their assumptions. Skip ahead to the "Ordinal Regression Models" section if you'd like.

1. I understand the limitations and advantages of AI writing tools like ChatGPT

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: value ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer

REML criterion at convergence: 1421.9

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.2093 -0.4810  0.2129  0.6655  2.0234

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 0.1993   0.4464
 Residual                   0.3479   0.5898
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                            Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                  1.22131    0.04735 572.11553  25.791  < 2e-
16 ***
timepost                     0.32377    0.05340 324.00000   6.063 3.71e-
09 ***
Term_prespring_25            0.16893    0.09442 572.11553   1.789   0.0741 .
timepost:Term_prespring_25  -0.21401    0.10647 324.00000  -2.010   0.0453 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.564
Trm_prsp_25 -0.502  0.283
tmpst:T__25  0.283 -0.502 -0.564

Type III Analysis of Variance Table with Satterthwaite's method
```

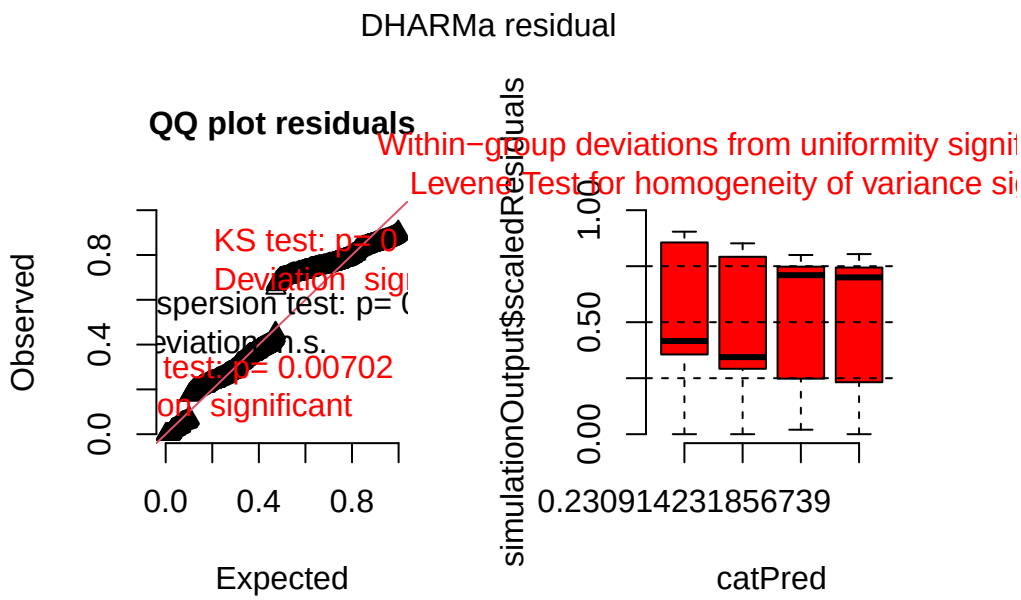
```
              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
time          5.7675  5.7675     1   324 16.5784 5.872e-05 ***
Term_pre      0.2194  0.2194     1   324  0.6306   0.42772
time:Term_pre 1.4055  1.4055     1   324  4.0401   0.04526 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

NOTE: Results may be misleading due to involvement in interactions

$emmeans
 time emmean     SE  df lower.CL upper.CL
 pre    1.31 0.0472 572     1.21     1.40
 post   1.52 0.0472 572     1.43     1.62

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast   estimate     SE  df t.ratio p.value
 pre - post   -0.217 0.0532 324  -4.072  0.0001

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger

NOTE: Results may be misleading due to involvement in interactions

$emmeans
 Term_pre  emmean     SE  df lower.CL upper.CL
 spring_24   1.38 0.0391 324     1.31     1.46
 spring_25   1.45 0.0675 324     1.31     1.58

Results are averaged over the levels of: time
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast              estimate    SE  df t.ratio p.value
 spring_24 - spring_25  -0.0619 0.078 324  -0.794  0.4277

Results are averaged over the levels of: time
Degrees-of-freedom method: kenward-roger
```

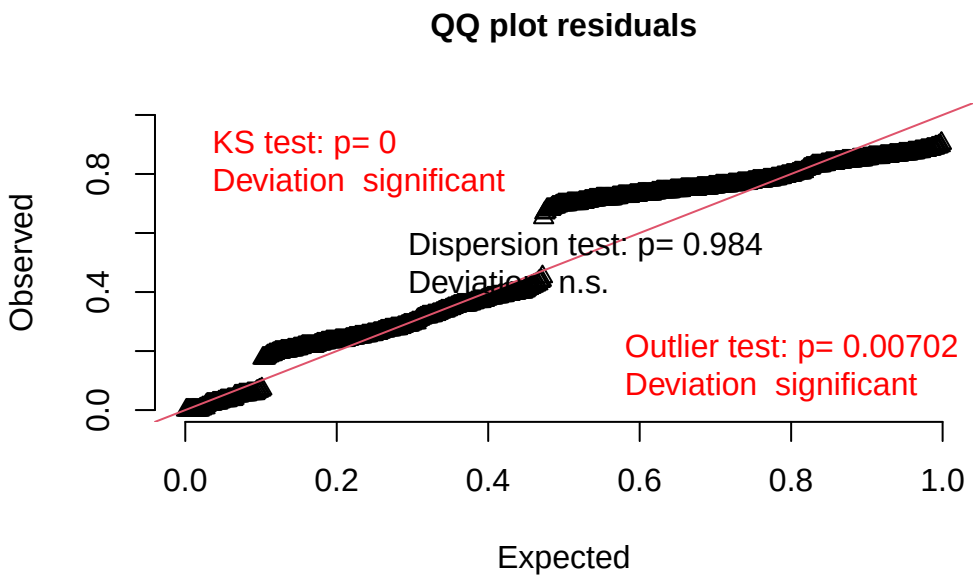
```
$emmeans
 time Term_pre  emmean     SE  df lower.CL upper.CL
 pre  spring_24   1.22 0.0474 572     1.13     1.31
 post spring_24   1.55 0.0474 572     1.45     1.64
 pre  spring_25   1.39 0.0817 572     1.23     1.55
 post spring_25   1.50 0.0817 572     1.34     1.66

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast                         estimate     SE  df t.ratio p.value
 pre spring_24 - post spring_24   -0.3238 0.0534 324  -6.063  <.0001
 pre spring_24 - pre spring_25    -0.1689 0.0944 572  -1.789  0.2796
 pre spring_24 - post spring_25   -0.2787 0.0944 572  -2.952  0.0173
 post spring_24 - pre spring_25    0.1548 0.0944 572   1.640  0.3570
 post spring_24 - post spring_25   0.0451 0.0944 572   0.477  0.9640
 pre spring_25 - post spring_25   -0.1098 0.0921 324  -1.192  0.6327

Degrees-of-freedom method: kenward-roger
P value adjustment: tukey method for comparing a family of 4 estimates


Ignoring unknown labels:
* linetype : "Term pre"
* shape : "Term pre"
```




```
# Effect Size for ANOVA (Type III)

Parameter     | Eta2 (partial) |       95% CI
---------------------------------------------
time          |           0.05 | [0.02, 1.00]
Term_pre      |       1.94e-03 | [0.00, 1.00]
time:Term_pre |           0.01 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

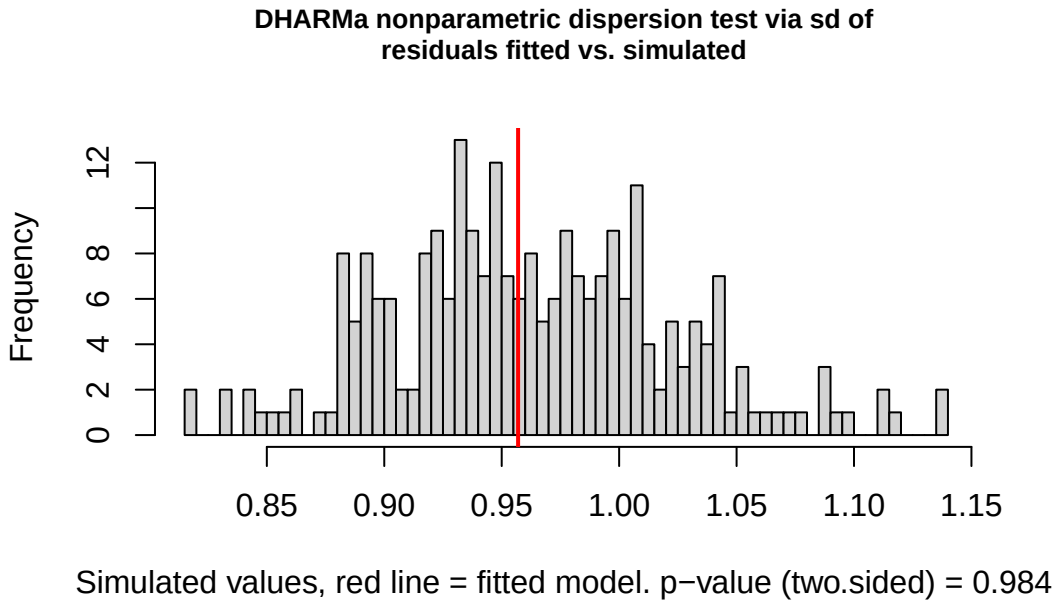
Checking for Assumptions & Model Diagnostics



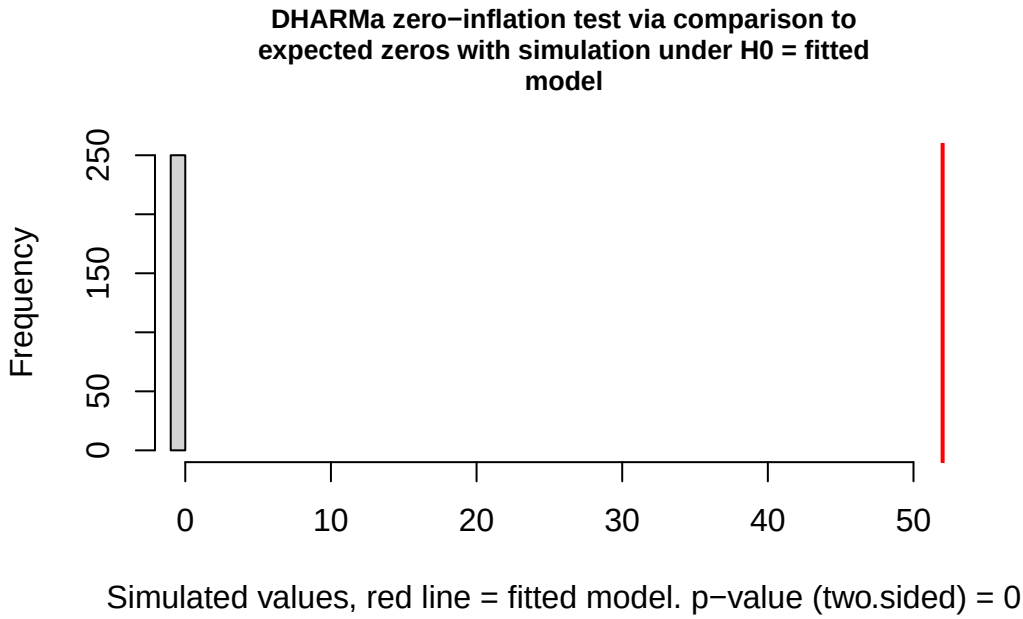



```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data:  simulationOutput$scaledResiduals
D = 0.20367, p-value < 2.2e-16
alternative hypothesis: two-sided
```




```
    DHARMa nonparametric dispersion test via sd of residuals fitted vs.
    simulated

data:  simulationOutput
dispersion = 0.99381, p-value = 0.984
alternative hypothesis: two.sided
```



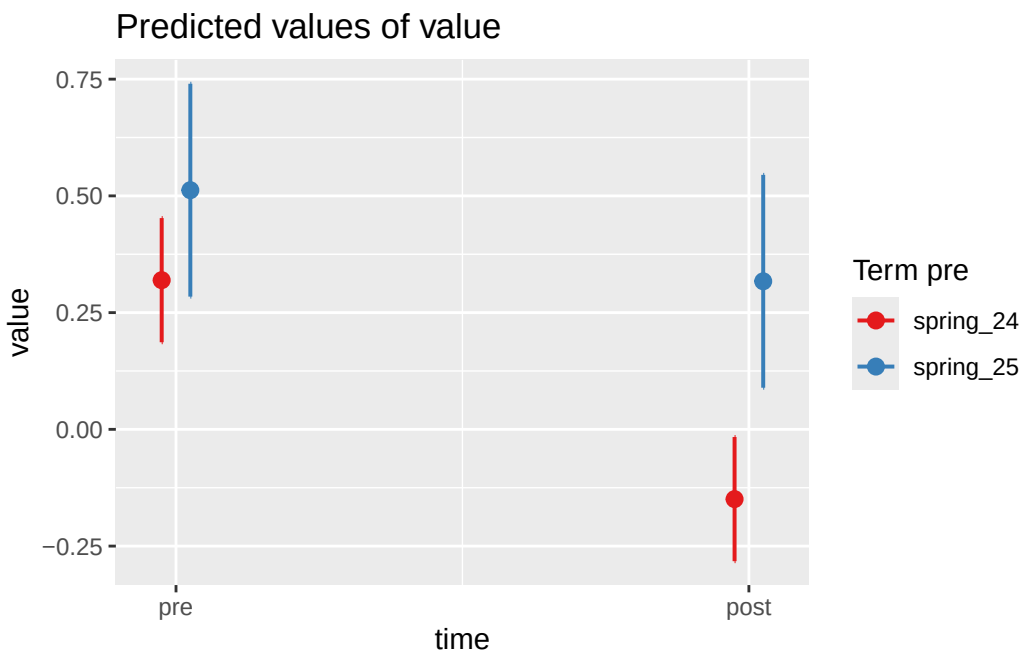

```
    DHARMa zero-inflation test via comparison to expected zeros with
    simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = Inf, p-value < 2.2e-16
alternative hypothesis: two.sided
```

2. Using an AI Writing Tool on a Writing Assignment is a form of Cheating/Plagiarism

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: value ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

REML criterion at convergence: 1865.1

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.37338 -0.62536 -0.07436  0.54978  2.27592

Random effects:
```

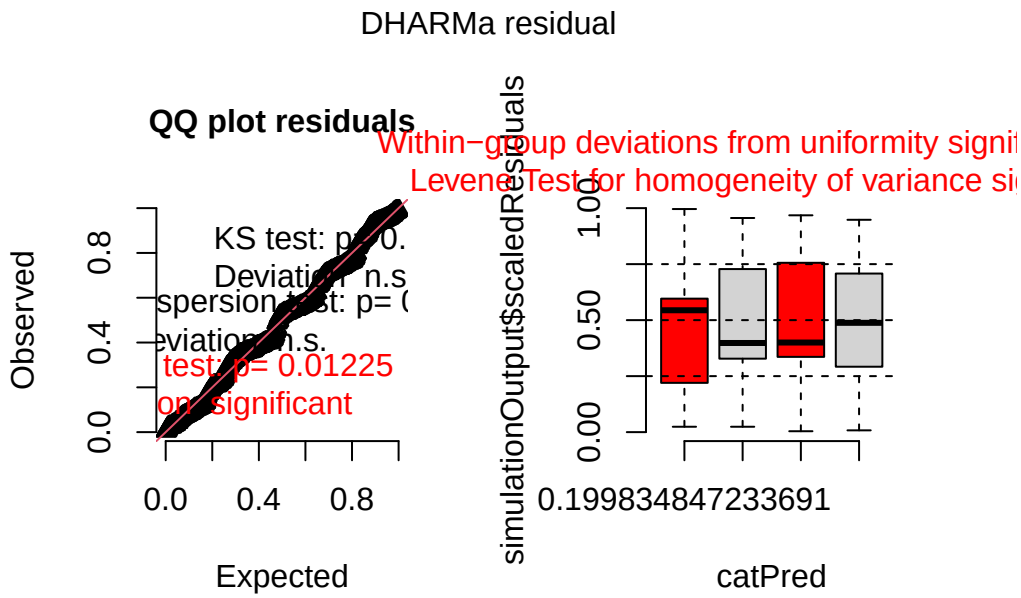
```
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 0.3806   0.617
 Residual                   0.7241   0.851
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
                           Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                 0.31950    0.06771 573.87641   4.719 2.98e-
06 ***
timepost                   -0.46888    0.07752 321.00000  -6.048 4.07e-
09 ***
Term_prespring_25           0.19269    0.13438 573.87641   1.434   0.1521
timepost:Term_prespring_25  0.27376    0.15385 321.00000   1.779   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.572
Trm_prsp_25 -0.504  0.288
tmpst:T__25  0.288 -0.504 -0.572

Type III Analysis of Variance Table with Satterthwaite's method
               Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
time          13.4877 13.4877     1   321 18.6259 2.121e-05 ***
Term_pre       6.4793  6.4793     1   321  8.9477  0.002993 **
time:Term_pre  2.2926  2.2926     1   321  3.1660  0.076132 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

NOTE: Results may be misleading due to involvement in interactions

$emmeans
 time emmean     SE  df lower.CL upper.CL
 pre  0.4158 0.0672 574   0.2839    0.548
 post 0.0838 0.0672 574  -0.0481    0.216

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
```

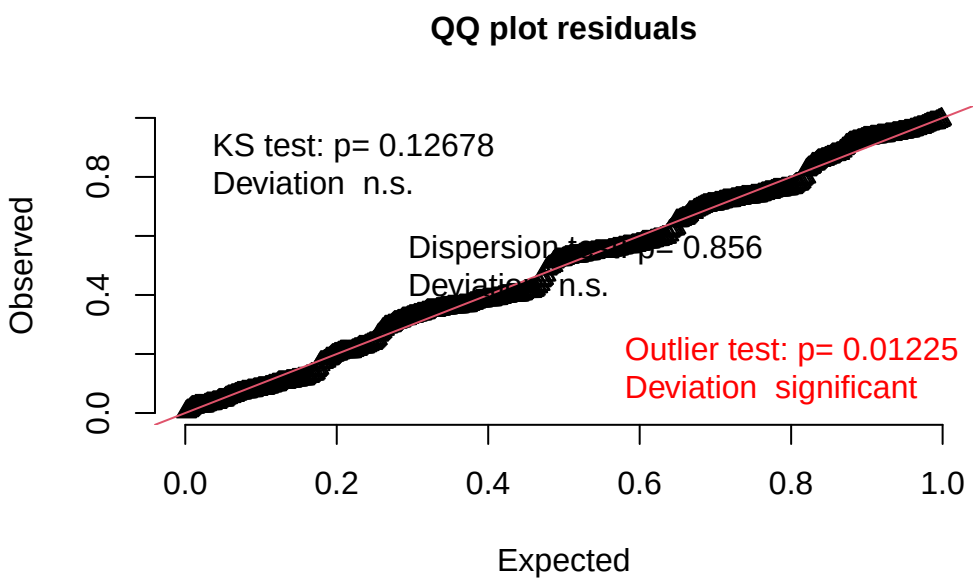
```
 contrast   estimate     SE  df t.ratio p.value
 pre - post    0.332 0.0769 321   4.316  <.0001

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger


Ignoring unknown labels:
* linetype : "Term pre"
* shape : "Term pre"
```




```
# Effect Size for ANOVA (Type III)

Parameter     | Eta2 (partial) |       95% CI
-----------------------------------------------
time          |           0.05 | [0.02, 1.00]
Term_pre      |           0.03 | [0.01, 1.00]
time:Term_pre |       9.77e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

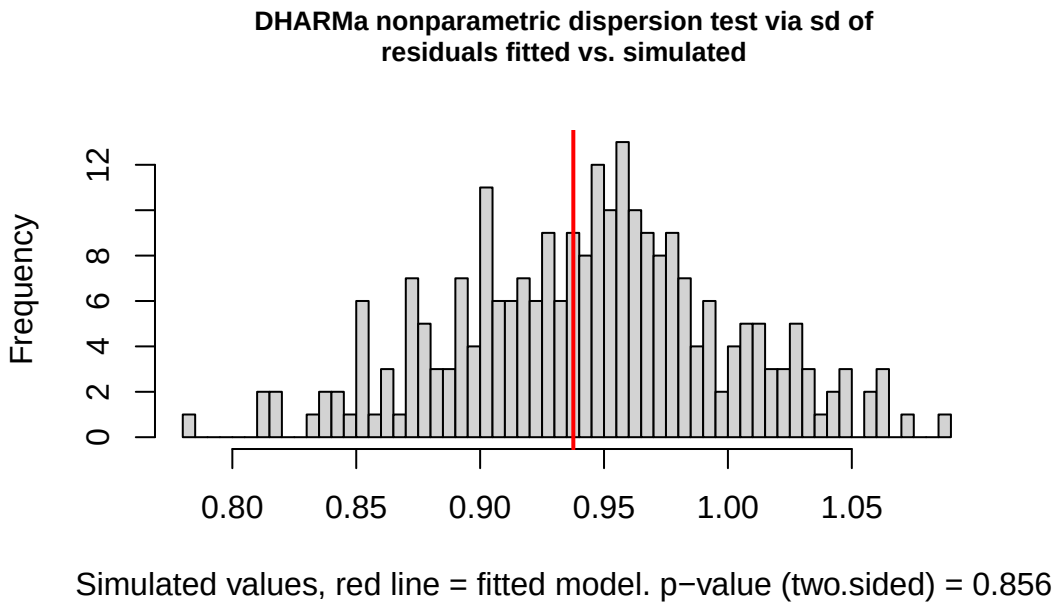
Checking for Assumptions & Model Diagnostics



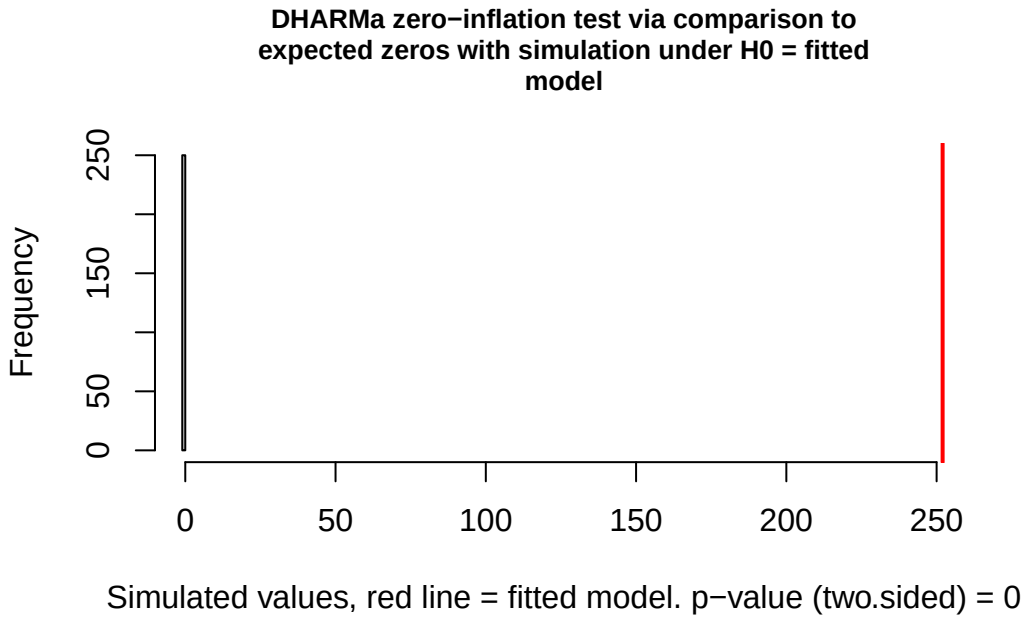



```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data:  simulationOutput$scaledResiduals
D = 0.046204, p-value = 0.1268
alternative hypothesis: two-sided
```




```
    DHARMa nonparametric dispersion test via sd of residuals fitted vs.
    simulated

data:  simulationOutput
dispersion = 0.99273, p-value = 0.856
alternative hypothesis: two.sided
```




```
    DHARMa zero-inflation test via comparison to expected zeros with
    simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = Inf, p-value < 2.2e-16
alternative hypothesis: two.sided
```

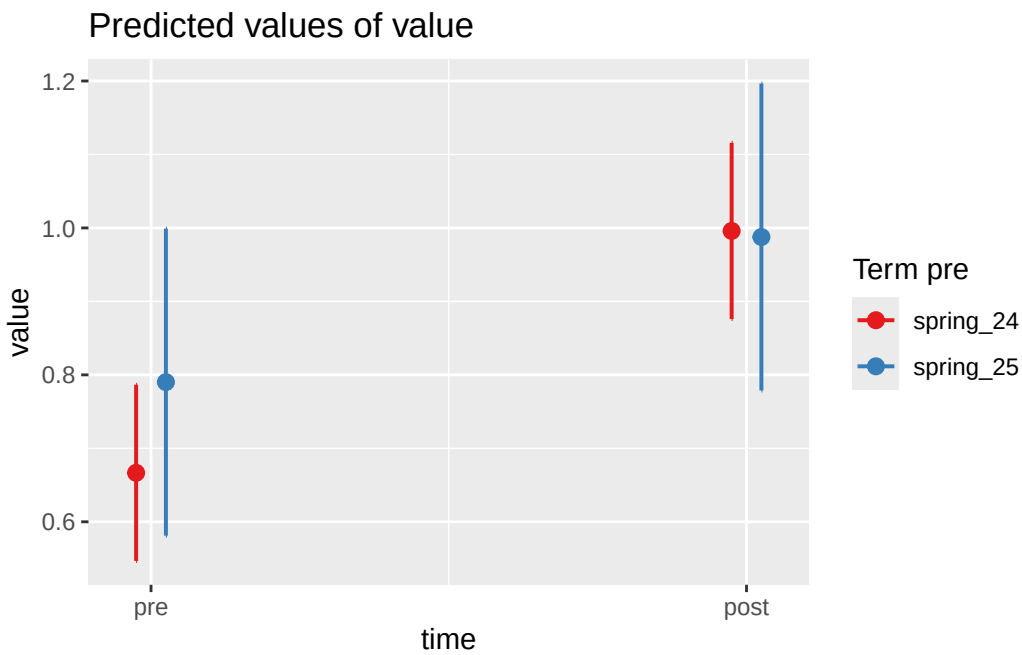
3. Peer-feedback on my writing is clear and/or helpful

```
Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]
Formula: value ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_peer_longer

REML criterion at convergence: 1768.9

Scaled residuals:
     Min       1Q   Median       3Q      Max
-3.00363 -0.42718  0.00583  0.54275  1.87313

Random effects:
```

```
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 0.3091   0.5560
 Residual                   0.6080   0.7797
Number of obs: 654, groups:  Participant_ID, 327

Fixed effects:
                           Estimate Std. Error        df t value Pr(>|t|)
(Intercept)                 0.66667    0.06106 583.68873  10.919  < 2e-
16 ***
timepost                    0.32927    0.07030 325.00000   4.683 4.15e-
06 ***
Term_prespring_25           0.12346    0.12268 583.68873   1.006    0.315
timepost:Term_prespring_25 -0.13174    0.14126 325.00000  -0.933    0.352
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.576
Trm_prsp_25 -0.498  0.287
tmpst:T__25  0.287 -0.498 -0.576

Type III Analysis of Variance Table with Satterthwaite's method
              Sum Sq Mean Sq NumDF DenDF F value    Pr(>F)
time          8.4554  8.4554     1   325 13.9080 0.0002265 ***
Term_pre      0.2004  0.2004     1   325  0.3296 0.5662751
time:Term_pre 0.5288  0.5288     1   325  0.8697 0.3517185
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

NOTE: Results may be misleading due to involvement in interactions

$emmeans
 time emmean     SE  df lower.CL upper.CL
 pre   0.728 0.0613 584    0.608    0.849
 post  0.992 0.0613 584    0.871    1.112

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
```

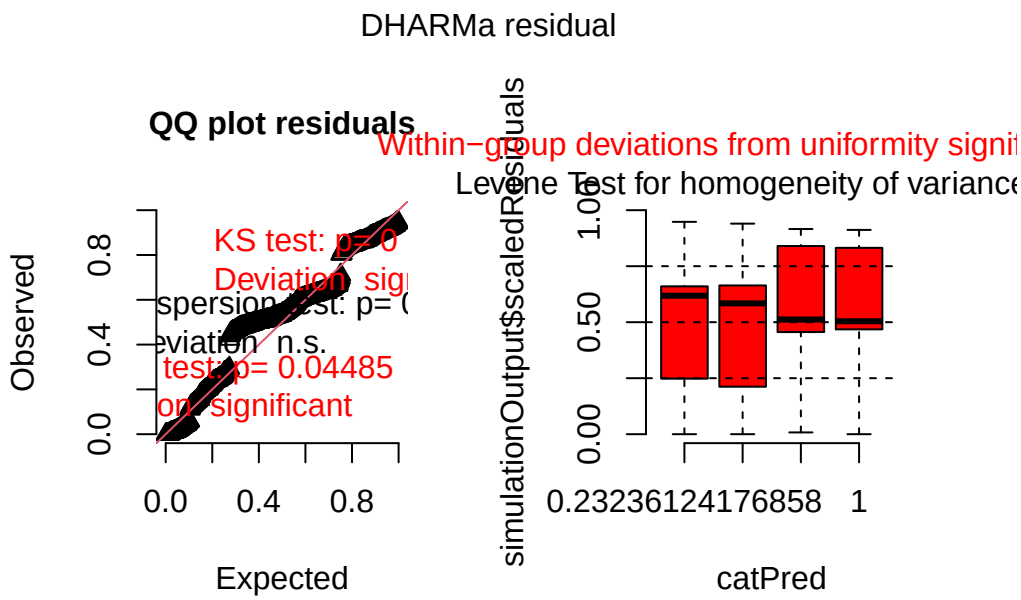
```
 contrast   estimate     SE  df t.ratio p.value
 pre - post   -0.263 0.0706 325  -3.729  0.0002

Results are averaged over the levels of: Term_pre
Degrees-of-freedom method: kenward-roger

NOTE: Results may be misleading due to involvement in interactions

$emmeans
 Term_pre  emmean     SE  df lower.CL upper.CL
 spring_24  0.831 0.0499 325    0.733     0.93
 spring_25  0.889 0.0870 325    0.718     1.06

Results are averaged over the levels of: time
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast              estimate  SE  df t.ratio p.value
 spring_24 - spring_25  -0.0576 0.1 325  -0.574  0.5663

Results are averaged over the levels of: time
Degrees-of-freedom method: kenward-roger

$emmeans
 time Term_pre  emmean     SE  df lower.CL upper.CL
 pre  spring_24  0.667 0.0611 584    0.547    0.787
 post spring_24  0.996 0.0611 584    0.876    1.116
 pre  spring_25  0.790 0.1060 584    0.581    0.999
 post spring_25  0.988 0.1060 584    0.779    1.197

Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

$contrasts
 contrast                        estimate     SE  df t.ratio p.value
 pre spring_24 - post spring_24  -0.32927 0.0703 325  -4.683  <.0001
 pre spring_24 - pre spring_25   -0.12346 0.1230 584  -1.006  0.7458
 pre spring_24 - post spring_25  -0.32099 0.1230 584  -2.617  0.0450
 post spring_24 - pre spring_25   0.20581 0.1230 584   1.678  0.3364
 post spring_24 - post spring_25  0.00828 0.1230 584   0.068  0.9999
 pre spring_25 - post spring_25  -0.19753 0.1230 325  -1.612  0.3731
```

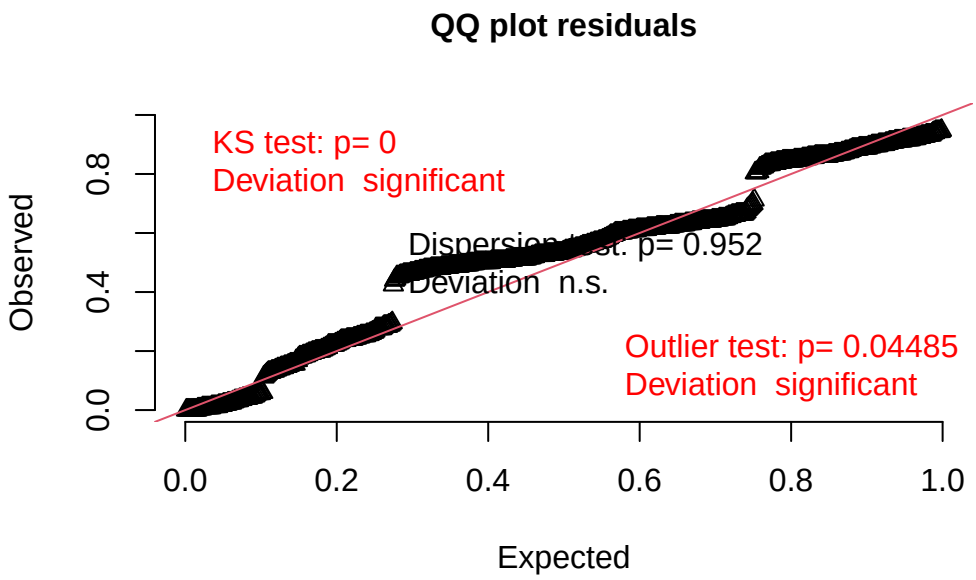
```
Degrees-of-freedom method: kenward-roger
P value adjustment: tukey method for comparing a family of 4 estimates

Ignoring unknown labels:
* linetype : "Term pre"
* shape : "Term pre"
```




```
# Effect Size for ANOVA (Type III)

Parameter     | Eta2 (partial) |       95% CI
-----------------------------------------------
time          |           0.04 | [0.01, 1.00]
Term_pre      |       1.01e-03 | [0.00, 1.00]
time:Term_pre |       2.67e-03 | [0.00, 1.00]

- One-sided CIs: upper bound fixed at [1.00].
```

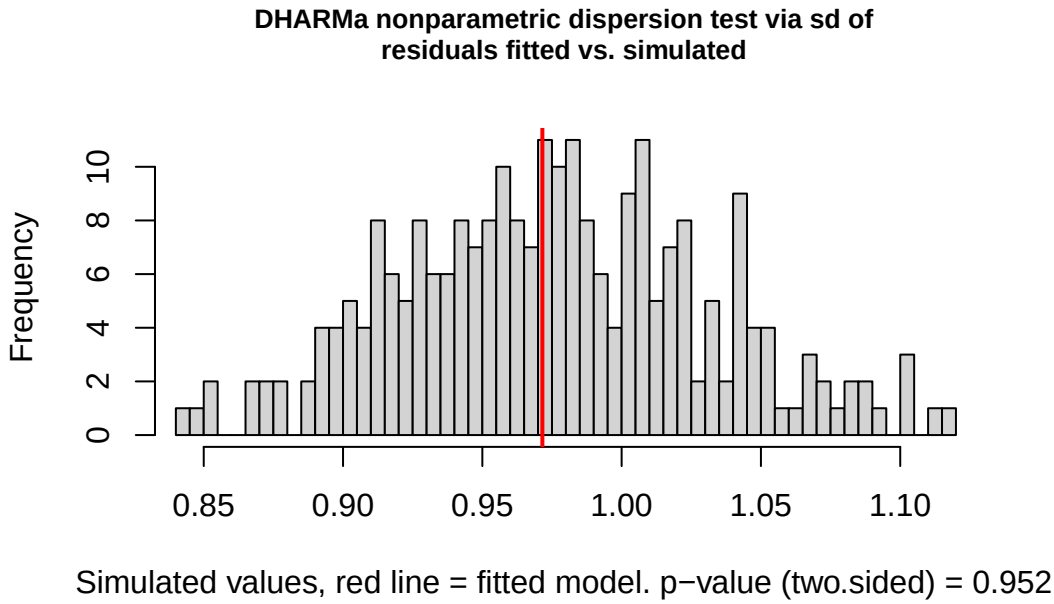
Checking for Assumptions & Model Diagnostics



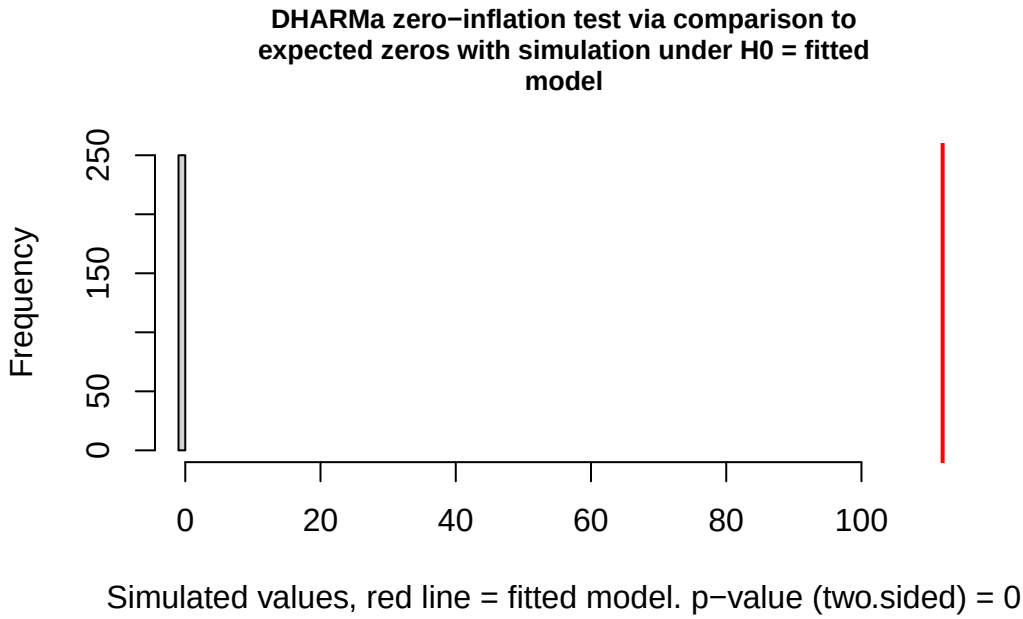



```
Asymptotic one-sample Kolmogorov-Smirnov test
```

```
data:  simulationOutput$scaledResiduals
D = 0.17371, p-value < 2.2e-16
alternative hypothesis: two-sided
```




```
    DHARMa nonparametric dispersion test via sd of residuals fitted vs.
    simulated

data:  simulationOutput
dispersion = 0.99644, p-value = 0.952
alternative hypothesis: two.sided
```




```
    DHARMa zero-inflation test via comparison to expected zeros with
    simulation under H0 = fitted model

data:  simulationOutput
ratioObsSim = Inf, p-value < 2.2e-16
alternative hypothesis: two.sided
```

Ordinal Regression Models

1) Using 'ordinal' package: https://cran.r-project.org/web/packages/ordinal/ordinal.pdf Cite: Christensen (Haubo Bojesen 2026)

This package implements cumualtive link models and cumulative link models with normally distributed random effects, denoted cumulative link mixed (effects) models. Cumulative link models are also known as ordered regression models, proportional odds models, proportional hazards models for grouped survival times and ordered logit/probit/… models.

Cumulative link mixed models are fitted with clmm function. Estimation is via maximum likelihood using the Laplace approximation or adaptive GaussHermite quadrature (one random effect).

Problem with 'ordinal' package is can't test for model assumptions using current most up-to-date function…

Tried GLMMAdaptive, rms, mass, and vgam packages. None have mixed effects models, except for GLMMAdaptive, but it didn't work for me.

```
[1] 9.5400e-08 4.2600e-06 4.2600e-06 1.6965e-01 2.4500e-01 2.4500e-01 2.4500e-
01
[8] 2.4500e-01 2.4500e-01
```

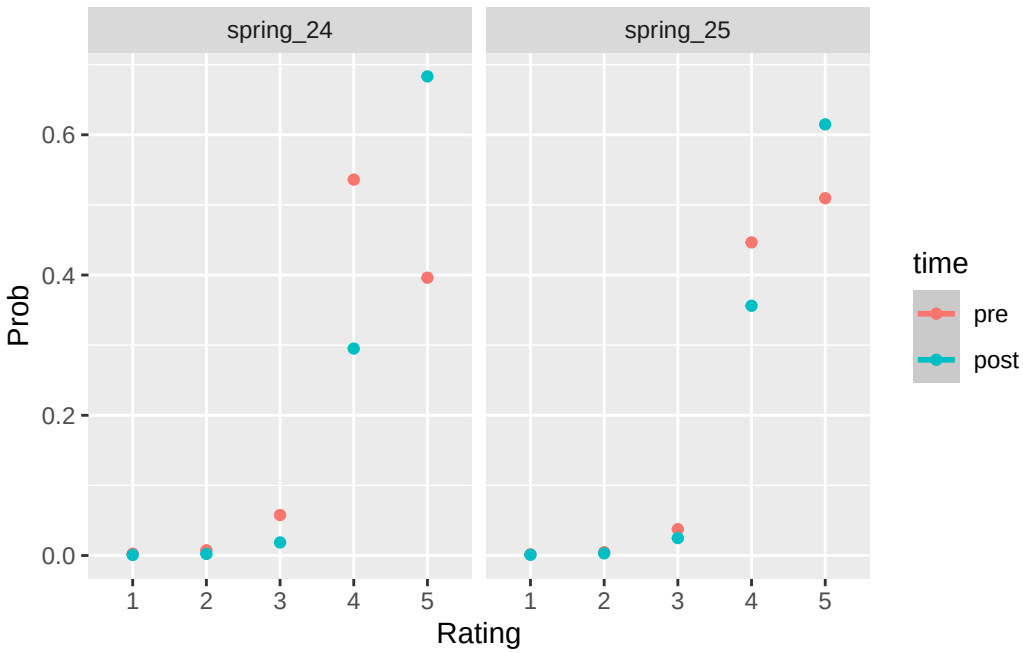
1. Limitations & Adv.: I understand the limitations and advantages of AI writing tools like ChatGPT

Model1: Interaction of time and term (i.e., year of data collection)

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: Value ~ time * Term_pre + (1 | Participant_ID)
data:    s_24_25_MIGS_072525_UL_longer

 link  threshold nobs logLik  AIC     niter     max.grad cond.H
 logit flexible  652  -618.63 1253.27 427(1692) 3.53e-04 7.5e+01

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 2.9      1.703
Number of groups:  Participant_ID 326

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                     1.1902     0.2199   5.413  6.2e-08 ***
Term_prespring_25            0.4597     0.3569   1.288   0.1977
timepost:Term_prespring_25  -0.7607     0.4031  -1.887   0.0591 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
      Estimate Std. Error z value
-2|-1  -5.9774     0.5892 -10.145
-1|0   -4.5975     0.3881 -11.847
0|1    -2.6220     0.2581 -10.157
1|2     0.4215     0.1893   2.226
```

```
Term_pre = spring_24:
 time emmean    SE  df asymp.LCL asymp.UCL
 pre    3.19 0.285 Inf      2.64      3.75
 post   4.38 0.345 Inf      3.71      5.06

Term_pre = spring_25:
 time emmean    SE  df asymp.LCL asymp.UCL
 pre    3.65 0.397 Inf      2.88      4.43
 post   4.08 0.417 Inf      3.27      4.90

Confidence level used: 0.95


   time  Term_pre Rating         Prob
1   pre spring_24      1 0.0025288739
2  post spring_24      1 0.0007705351
3   pre spring_25      1 0.0015983641
4  post spring_25      1 0.0010408825
5   pre spring_24      2 0.0074476947
6  post spring_24      2 0.0022851386
7   pre spring_25      2 0.0047246557
8  post spring_25      2 0.0030835855
9   pre spring_24      3 0.0577591436
10 post spring_24      3 0.0185658308
11  pre spring_25      3 0.0375442336
12 post spring_25      3 0.0248708022
13  pre spring_24      4 0.5361059896
14 post spring_24      4 0.2951373266
15  pre spring_25      4 0.4465769983
16 post spring_25      4 0.3561660619
17  pre spring_24      5 0.3961582983
18 post spring_24      5 0.6832411688
19  pre spring_25      5 0.5095557482
20 post spring_25      5 0.6148386678


   time  Term_pre Rating      Cumprob
1   pre spring_24      1 0.0025288739
2  post spring_24      1 0.0007705351
3   pre spring_25      1 0.0015983641
4  post spring_25      1 0.0010408825
5   pre spring_24      2 0.0099765686
6  post spring_24      2 0.0030556738
7   pre spring_25      2 0.0063230199
```

```
8  post spring_25      2 0.0041244680
9   pre spring_24      3 0.0677357122
10 post spring_24      3 0.0216215046
11  pre spring_25      3 0.0438672535
12 post spring_25      3 0.0289952702
13  pre spring_24      4 0.6038417017
14 post spring_24      4 0.3167588312
15  pre spring_25      4 0.4904442518
16 post spring_25      4 0.3851613322
17  pre spring_24      5 1.0000000000
18 post spring_24      5 1.0000000000
19  pre spring_25      5 1.0000000000
20 post spring_25      5 1.0000000000

  time  Term_pre Rating
1  pre spring_24      4
2 post spring_24      5
3  pre spring_25      5
4 post spring_25      5

`geom_smooth()` using formula = 'y ~ x'
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
`geom_line()`: Each group consists of only one observation.
i Do you need to adjust the group aesthetic?
```




```
$Participant_ID
            (Intercept)
(Intercept)    2.900032
attr(,"stddev")
(Intercept)
   1.702948


$Models

Model: "clmm, Value ~ time * Term_pre + (1 | Participant_ID), s_24_25_MIGS_072525_UL_longer"
Null:  "clmm, Value ~ 1 + (1 | Participant_ID), s_24_25_MIGS_072525_UL_longer"

$Pseudo.R.squared.for.model.vs.null
                             Pseudo.R.squared
McFadden                            0.0262091
Cox and Snell (ML)                  0.0497919
Nagelkerke (Cragg and Uhler)        0.0580633

$Likelihood.ratio.test
 Df.diff LogLik.diff Chisq    p.value
      -3      -16.65  33.3 2.7834e-07
```

```
$Number.of.observations

Model: 652
Null:  652

$Messages
[1] "Note: For models fit with REML, these statistics are based on refitting with ML"

$Warnings
[1] "None"
```

Time is a predictor of values. Term is not a predictor of values, and there's an almost-significant interaction between time and term. Was sig. in LMER model.

Checking for Assumptions & Model Diagnostics

1. **Correct link & functional form:** The chosen link (logit/probit/cloglog/cauchit) appropriately relates predictors to the latent continuous variable. Predictors enter the model with an appropriate functional form (e.g., linear on the latent scale unless modeled otherwise) [this is good!]
2. **Proportional odds assumption:** The effect of the explanatory variables remains constant across all categories j. Violating this assumption can lead to model misspecification. Several classical methods exist for testing the assumption, including the Brant, Score, and Wald tests [22]. Within the probabilistic framework, one can evaluate the assumption using leave-one-out cross-validation (LOO-CV). This approach involves comparing the cumulative model to a more flexible alternative that allows category-specific effects. If the alternative model yields a better fit, it indicates that the proportional odds assumption may not hold [4]. / The effect of each predictor is constant across cumulative thresholds (unless you relax this via "nominal" effects). If a predictor increases the chance of going from "low → medium", it should increase the chance of "medium → high" by a similar amount. If not, the model might be *oversimplifying* how your predictors work.
3. **Random-effects structure is appropriate:** Random effects capture grouping; their **conditional modes** are approximately normal, and the variance components are meaningful (not spuriously ~0).
4. **No severe multicollinearity or separation:** Predictors are not collinear; there's no (quasi-)separation that causes estimation problems.
5. **Overall fit / calibration:** Predicted cumulative probabilities are consistent with observed category frequencies.

Brand Test: Proportional Odds Assumption

```
------------------------------------------------------------
Test for            X2  df  probability
------------------------------------------------------------
Omnibus             9.27    9   0.41
timepost            4.24    3   0.24
Term_prespring_25       4   3   0.26
timepost:Term_prespring_25  0.4 3   0.94
------------------------------------------------------------

H0: Parallel Regression Assumption holds

Warning in brant::brant(polr_fit): 4 combinations in table(dv,ivs) do not
occur. Because of that, the test results might be invalid.
```

Error message:

```
Error in `optim()`:
! initial value in 'vmmin' is not finite
```

Informal checks with GLMM– not sure that doing correctly

```
Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
unable to evaluate scaled gradient

Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
Hessian is numerically singular: parameters are not uniquely determined

Warning in vcov.merMod(object, use.hessian = use.hessian): variance-covariance matrix compute
difference Hessian is
not positive definite or contains NA values: falling back to var-cov estimated from RX

Warning in vcov.merMod(object, correlation = correlation, sigm = sig): variance-
covariance matrix computed from finite-difference Hessian is
not positive definite or contains NA values: falling back to var-cov estimated from RX

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == -2) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer
```

```
      AIC       BIC    logLik -2*log(L)  df.resid
     22.9      45.3      -6.5      12.9       647

Scaled residuals:
      Min        1Q    Median        3Q       Max
-0.000449 -0.000449  0.000000  0.000000  0.069437

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 4324     65.76
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -1.542e+01  3.278e+01  -0.470    0.638
timepost                   -7.842e+04  4.296e+06  -0.018    0.985
Term_prespring_25          -5.581e+03  7.411e+06  -0.001    0.999
timepost:Term_prespring_25  1.776e+04  1.133e+07   0.002    0.999

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost     0.000
Trm_prsp_25  0.000  0.000
tmpst:T__25  0.000 -0.379 -0.654
optimizer (Nelder_Mead) convergence code: 0 (OK)
unable to evaluate scaled gradient
 Hessian is numerically singular: parameters are not uniquely determined

Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
unable to evaluate scaled gradient

Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, :
Hessian is numerically singular: parameters are not uniquely determined

Warning in vcov.merMod(object, use.hessian = use.hessian): variance-covariance matrix compute
difference Hessian is
not positive definite or contains NA values: falling back to var-cov estimated from RX

Warning in vcov.merMod(object, correlation = correlation, sigm = sig): variance-
covariance matrix computed from finite-difference Hessian is
not positive definite or contains NA values: falling back to var-cov estimated from RX
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == -1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer

      AIC       BIC    logLik -2*log(L)  df.resid
     60.5      82.9     -25.2      50.5       647

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.3474 -0.0030 -0.0001 -0.0001  8.9887

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 382      19.55
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                             Estimate Std. Error z value Pr(>|z|)
(Intercept)                -1.159e+01  6.787e+00  -1.707   0.0877 .
timepost                   -7.331e+00  3.805e+00  -1.927   0.0540 .
Term_prespring_25          -7.525e-01  2.006e+01  -0.038   0.9701
timepost:Term_prespring_25 -1.417e+03  7.411e+06   0.000   0.9998
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.152
Trm_prsp_25 -0.338  0.052
tmpst:T__25  0.000  0.000  0.000
optimizer (Nelder_Mead) convergence code: 0 (OK)
unable to evaluate scaled gradient
 Hessian is numerically singular: parameters are not uniquely determined


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == 0) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer
```

```
      AIC       BIC    logLik -2*log(L)  df.resid
    328.7     351.1    -159.3     318.7       647

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.74181 -0.03067 -0.01316 -0.00637  2.76303

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 60.72    7.792
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -6.9097     0.7907  -8.739  < 2e-16 ***
timepost                    -3.1425     0.7802  -4.028 5.63e-05 ***
Term_prespring_25           -1.0471     1.1130  -0.941   0.3468
timepost:Term_prespring_25   2.4692     1.3901   1.776   0.0757 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost     0.382
Trm_prsp_25 -0.223  0.050
tmpst:T__25 -0.171 -0.533 -0.344


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value <= 1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    857.0     879.4    -423.5     847.0       647

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.2340 -0.6241 -0.3750  0.6954  1.3489

Random effects:
 Groups         Name        Variance Std.Dev.
```

```
 Participant_ID (Intercept) 3.062    1.75
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                  0.2927     0.2016   1.452    0.146
timepost                    -1.0190     0.2470  -4.125 3.71e-05 ***
Term_prespring_25           -0.1338     0.4003  -0.334    0.738
timepost:Term_prespring_25   0.5412     0.4652   1.163    0.245
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.573
Trm_prsp_25 -0.501  0.279
tmpst:T__25  0.296 -0.505 -0.575


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value < 2) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_UL_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    857.0     879.4    -423.5     847.0       647

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.2340 -0.6241 -0.3750  0.6954  1.3489

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 3.062    1.75
Number of obs: 652, groups:  Participant_ID, 326

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                  0.2927     0.2016   1.452    0.146
timepost                    -1.0190     0.2470  -4.125 3.71e-05 ***
Term_prespring_25           -0.1338     0.4003  -0.334    0.738
timepost:Term_prespring_25   0.5412     0.4652   1.163    0.245
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.573
Trm_prsp_25 -0.501  0.279
tmpst:T__25  0.296 -0.505 -0.575
```

Checking other assumptions

```
formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_UL_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  652  -644.06 1302.11 6(0)  2.58e-11 7.2e+01

Coefficients:
                  timepost          Term_prespring_25
                    0.7518                     0.2643
timepost:Term_prespring_25
                   -0.4694

Threshold coefficients:
  -2|-1    -1|0     0|1     1|2
-4.7568 -3.4843 -1.8338  0.2783

formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_UL_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  652  -644.06 1302.11 6(0)  2.58e-11 7.2e+01

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                     0.7518     0.1793   4.194 2.74e-05 ***
Term_prespring_25            0.2643     0.2418   1.093    0.274
timepost:Term_prespring_25  -0.4694     0.3468  -1.353    0.176
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
```

```
      Estimate Std. Error z value
-2|-1  -4.7568     0.5084  -9.356
-1|0   -3.4843     0.2831 -12.307
0|1    -1.8338     0.1575 -11.646
1|2     0.2783     0.1276   2.181

Tests of scale effects

formula: Value ~ time * Term_pre
              Df  logLik    AIC     LRT Pr(>Chi)
<none>           -644.06 1302.1
time           1 -641.22 1298.4  5.6773 0.017186 *
Term_pre       1 -640.87 1297.7  6.3690 0.011613 *
time:Term_pre  3 -638.11 1296.2 11.8979 0.007741 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Tests of nominal effects

formula: Value ~ time * Term_pre
              Df  logLik    AIC    LRT Pr(>Chi)
<none>           -644.06 1302.1
time           3 -640.66 1301.3 6.7957  0.07870 .
Term_pre       3 -640.65 1301.3 6.8046  0.07839 .
time:Term_pre
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio tests of cumulative link models:

       formula:                                        link: threshold:
UL_fm  Value ~ time * Term_pre                         logit flexible
UL_fmm Value ~ time * Term_pre + (1 | Participant_ID) logit flexible

       no.par    AIC  logLik LR.stat df Pr(>Chisq)
UL_fm       7 1302.1 -644.06
UL_fmm      8 1253.3 -618.63  50.848  1  9.981e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

[1] 1253.265
```

```
[1] 1433.905
```

spring_24

Descriptive Statistics (N=488)

	pre (N=244)	post (N=244)
Value : -2	0.02 (4)	0.00 (0)
-1	0.03 (7)	0.01 (2)
0	0.13 (31)	0.05 (12)
1	0.37 (91)	0.33 (81)
2	0.45 (111)	0.61 (149)

spring_25

Descriptive Statistics (N=164)

	pre (N=82)	post (N=82)
Value : -2	0.00 (0)	0.00 (0)
-1	0.01 (1)	0.00 (0)
0	0.06 (5)	0.05 (4)
1	0.45 (37)	0.40 (33)
2	0.48 (39)	0.55 (45)

```
# Check for Multicollinearity

Low Correlation

          Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
          time 2.02 [1.82, 2.28]     1.42      0.49     [0.44, 0.55]
      Term_pre 3.08 [2.73, 3.51]     1.76      0.32     [0.28, 0.37]
 time:Term_pre 2.36 [2.11, 2.67]     1.54      0.42     [0.37, 0.47]
```

2. Plagiarism: Using an AI Writing Tool on a Writing Assignment is a form of Cheating/Plagiarism

Model1: Interaction of time and term (i.e., year of data collection)

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: Value ~ time * Term_pre + (1 | Participant_ID)
data:    s_24_25_MIGS_072525_Pl_longer

 link  threshold nobs logLik  AIC     niter     max.grad cond.H
 logit flexible  646  -904.99 1825.99 509(1529) 7.27e-05 4.0e+01

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 1.862    1.364
Number of groups:  Participant_ID 323

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                    -1.0435     0.1824  -5.721 1.06e-08 ***
Term_prespring_25            0.4289     0.2996   1.431   0.1524
timepost:Term_prespring_25   0.6096     0.3428   1.778   0.0754 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
      Estimate Std. Error z value
-2|-1  -3.9384     0.2848 -13.829
-1|0   -1.8109     0.1910  -9.481
0|1     0.5156     0.1641   3.141
1|2     2.2156     0.2074  10.681
```

```
$Participant_ID
            (Intercept)
(Intercept)    1.861851
attr(,"stddev")
(Intercept)
   1.364497

   time  Term_pre Rating       Prob
1   pre spring_24      1 0.01910670
2  post spring_24      1 0.05240429
3   pre spring_25      1 0.01252681
4  post spring_25      1 0.01920171
5   pre spring_24      2 0.12142557
6  post spring_24      2 0.26463948
7   pre spring_25      2 0.08371223
8  post spring_25      2 0.12194247
9   pre spring_24      3 0.48558358
10 post spring_24      3 0.50917676
11  pre spring_25      3 0.42543131
12 post spring_25      3 0.48615473
13  pre spring_24      4 0.27552896
14 post spring_24      4 0.13677894
15  pre spring_25      4 0.33486190
16 post spring_25      4 0.27479346
17  pre spring_24      5 0.09835519
18 post spring_24      5 0.03700053
19  pre spring_25      5 0.14346776
20 post spring_25      5 0.09790763

   time  Term_pre Rating    Cumprob
1   pre spring_24      1 0.01910670
2  post spring_24      1 0.05240429
3   pre spring_25      1 0.01252681
4  post spring_25      1 0.01920171
5   pre spring_24      2 0.14053227
6  post spring_24      2 0.31704377
7   pre spring_25      2 0.09623903
8  post spring_25      2 0.14114418
9   pre spring_24      3 0.62611585
10 post spring_24      3 0.82622053
11  pre spring_25      3 0.52167034
12 post spring_25      3 0.62729891
```

```
13  pre spring_24      4 0.90164481
14 post spring_24      4 0.96299947
15  pre spring_25      4 0.85653224
16 post spring_25      4 0.90209237
17  pre spring_24      5 1.00000000
18 post spring_24      5 1.00000000
19  pre spring_25      5 1.00000000
20 post spring_25      5 1.00000000

  time  Term_pre Rating
1  pre spring_24      3
2 post spring_24      3
3  pre spring_25      3
4 post spring_25      3

$Models

Model: "clmm, Value ~ time * Term_pre + (1 | Participant_ID), s_24_25_MIGS_072525_Pl_longer"
Null:  "clmm, Value ~ 1 + (1 | Participant_ID), s_24_25_MIGS_072525_Pl_longer"

$Pseudo.R.squared.for.model.vs.null
                             Pseudo.R.squared
McFadden                            0.0243554
Cox and Snell (ML)                  0.0675534
Nagelkerke (Cragg and Uhler)        0.0716062

$Likelihood.ratio.test
 Df.diff LogLik.diff  Chisq    p.value
      -3     -22.592 45.183 8.4582e-10

$Number.of.observations

Model: 646
Null:  646

$Messages
[1] "Note: For models fit with REML, these statistics are based on refitting with ML"

$Warnings
[1] "None"
```

Significant for time, almost for interaction.

Checking for Assumptions & Model Diagnostics

Brand Test: Proportional Odds Assumption

```
------------------------------------------------------------
Test for            X2  df  probability
------------------------------------------------------------
Omnibus             12.72   9   0.18
timepost            6.6 3   0.09
Term_prespring_25       1.26    3   0.74
timepost:Term_prespring_25  1.73    3   0.63
------------------------------------------------------------

H0: Parallel Regression Assumption holds
```

Informal checks with GLMM– not sure that doing correctly

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == -2) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    215.7     238.0    -102.8     205.7       641

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.5431 -0.0137 -0.0037 -0.0014  3.9397

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 141.9    11.91
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -13.159      1.801  -7.305 2.77e-13 ***
timepost                      4.609      1.154   3.994 6.50e-05 ***
Term_prespring_25             1.952      3.005   0.650   0.5159
timepost:Term_prespring_25   -4.609      2.721  -1.694   0.0903 .
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.875
Trm_prsp_25 -0.380  0.356
tmpst:T__25  0.371 -0.424 -0.522


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == -1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    631.7     654.1    -310.9     621.7       641

Scaled residuals:
    Min      1Q  Median      3Q     Max
-0.6654 -0.3357 -0.2788 -0.2727  1.8434

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 2.163    1.471
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                -2.17940    0.30894  -7.055 1.73e-12 ***
timepost                    0.37153    0.26234   1.416    0.157
Term_prespring_25          -0.05273    0.43603  -0.121    0.904
timepost:Term_prespring_25  0.03406    0.52200   0.065    0.948
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.529
Trm_prsp_25 -0.330  0.317
tmpst:T__25  0.220 -0.496 -0.648


Generalized linear mixed model fit by maximum likelihood (Laplace
```

```
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value == 0) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    859.8     882.2    -424.9     849.8       641

Scaled residuals:
    Min      1Q  Median      3Q     Max
-0.9631 -0.7411 -0.6028  1.0384  1.4275

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 0.5095   0.7138
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -0.6829     0.1550  -4.407 1.05e-05 ***
timepost                     0.5240     0.2002   2.617  0.00887 **
Term_prespring_25           -0.1817     0.3023  -0.601  0.54786
timepost:Term_prespring_25  -0.3402     0.4020  -0.846  0.39737
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.694
Trm_prsp_25 -0.467  0.328
tmpst:T__25  0.339 -0.494 -0.679


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value <= 1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    440.0     462.3    -215.0     430.0       641

Scaled residuals:
```

```
     Min       1Q   Median       3Q      Max
-2.41036  0.00676  0.01431  0.02812  1.72610

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 78.21    8.844
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                  7.0774     0.7319   9.670  < 2e-16 ***
timepost                     2.8513     0.7280   3.917 8.98e-05 ***
Term_prespring_25           -0.3239     0.8319  -0.389    0.697
timepost:Term_prespring_25  -1.2097     1.0927  -1.107    0.268
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost     0.301
Trm_prsp_25 -0.318  0.077
tmpst:T__25 -0.054 -0.577 -0.222


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value < 2) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_Pl_longer

      AIC       BIC    logLik -2*log(L)  df.resid
    440.0     462.3    -215.0     430.0       641

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.41036  0.00676  0.01431  0.02812  1.72610

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 78.21    8.844
Number of obs: 646, groups:  Participant_ID, 323

Fixed effects:
```

```
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                  7.0774     0.7319   9.670  < 2e-16 ***
timepost                     2.8513     0.7280   3.917 8.98e-05 ***
Term_prespring_25           -0.3239     0.8319  -0.389    0.697
timepost:Term_prespring_25  -1.2097     1.0927  -1.107    0.268
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost     0.301
Trm_prsp_25 -0.318  0.077
tmpst:T__25 -0.054 -0.577 -0.222
```

Checking other assumptions

```
formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_Pl_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  646  -924.21 1862.41 5(0)  2.72e-08 4.9e+01

Coefficients:
                  timepost          Term_prespring_25
                   -0.8050                     0.3103
timepost:Term_prespring_25
                    0.4540

Threshold coefficients:
  -2|-1    -1|0     0|1     1|2
-3.0871 -1.3766  0.3603  1.6728

formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_Pl_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  646  -924.21 1862.41 5(0)  2.72e-08 4.9e+01

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                    -0.8050     0.1680  -4.793 1.64e-06 ***
Term_prespring_25            0.3103     0.2325   1.335    0.182
```

```
timepost:Term_prespring_25   0.4540     0.3277   1.385    0.166
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
      Estimate Std. Error z value
-2|-1  -3.0871     0.1962 -15.732
-1|0   -1.3766     0.1356 -10.152
0|1     0.3603     0.1231   2.927
1|2     1.6728     0.1457  11.484

Tests of scale effects

formula: Value ~ time * Term_pre
              Df  logLik    AIC     LRT Pr(>Chi)
<none>           -924.21 1862.4
time           1 -924.01 1864.0 0.39730   0.5285
Term_pre       1 -924.13 1864.3 0.15614   0.6927
time:Term_pre  3 -923.94 1867.9 0.53289   0.9116

Tests of nominal effects

formula: Value ~ time * Term_pre
              Df  logLik    AIC    LRT Pr(>Chi)
<none>           -924.21 1862.4
time           3 -921.64 1863.3 5.1413  0.16173
Term_pre       3 -919.59 1859.2 9.2230  0.02647 *
time:Term_pre
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio tests of cumulative link models:

       formula:                                        link: threshold:
Pl_fm  Value ~ time * Term_pre                         logit flexible
Pl_fmm Value ~ time * Term_pre + (1 | Participant_ID) logit flexible

       no.par    AIC  logLik LR.stat df Pr(>Chisq)
Pl_fm       7 1862.4 -924.21
Pl_fmm      8 1826.0 -904.99  38.426  1  5.686e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
[1] 1825.986

[1] 1877.106
```

spring_24

Descriptive Statistics (N=482)

	pre (N=241)	post (N=241)
Value : -2	0.05 (11)	0.11 (26)
-1	0.17 (42)	0.22 (53)
0	0.35 (85)	0.46 (112)
1	0.27 (65)	0.13 (31)
2	0.16 (38)	0.08 (19)

spring_25

Descriptive Statistics (N=164)

	pre (N=82)	post (N=82)
Value : -2	0.01 (1)	0.01 (1)
-1	0.17 (14)	0.22 (18)
0	0.32 (26)	0.35 (29)
1	0.29 (24)	0.27 (22)

```
|    2     |0.21  (17)|0.15  (12)|
+----------+----------+----------+


# Check for Multicollinearity

Low Correlation

          Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
          time 2.90 [2.57, 3.30]     1.70      0.35     [0.30, 0.39]
      Term_pre 3.25 [2.86, 3.70]     1.80      0.31     [0.27, 0.35]
 time:Term_pre 2.74 [2.43, 3.11]     1.65      0.37     [0.32, 0.41]
```

3. Peer-feedback on my writing is clear and/or helpful

Model1: Interaction of time and term (i.e., year of data collection)

```
Cumulative Link Mixed Model fitted with the Laplace approximation

formula: Value ~ time * Term_pre + (1 | Participant_ID)
data:    s_24_25_MIGS_072525_peer_longer

 link  threshold nobs logLik  AIC     niter     max.grad cond.H
 logit flexible  654  -818.91 1653.83 471(1792) 7.49e-05 4.1e+01

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 1.701    1.304
Number of groups:  Participant_ID 327

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                     0.8794     0.1824   4.823 1.42e-06 ***
Term_prespring_25            0.3522     0.3027   1.163    0.245
timepost:Term_prespring_25  -0.4366     0.3570  -1.223    0.221
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
      Estimate Std. Error z value
-2|-1  -4.0219     0.3261 -12.333
-1|0   -2.3649     0.2139 -11.054
```

```
0|1    -0.8499     0.1654  -5.139
1|2     1.9853     0.1945  10.209

$Participant_ID
            (Intercept)
(Intercept)    1.701022
attr(,"stddev")
(Intercept)
   1.304232

   time  Term_pre Rating        Prob
1   pre spring_24      1 0.017603242
2  post spring_24      1 0.007381687
3   pre spring_25      1 0.012442577
4  post spring_25      1 0.008026454
5   pre spring_24      2 0.068281603
6  post spring_24      2 0.030147638
7   pre spring_25      2 0.049526498
8  post spring_25      2 0.032672964
9   pre spring_24      3 0.213562144
10 post spring_24      3 0.113139569
11  pre spring_25      3 0.169127281
12 post spring_25      3 0.121089949
13  pre spring_24      4 0.579798321
14 post spring_24      4 0.600689645
15  pre spring_25      4 0.605498128
16 post spring_25      4 0.604998428
17  pre spring_24      5 0.120754690
18 post spring_24      5 0.248641460
19  pre spring_25      5 0.163405515
20 post spring_25      5 0.233212205

   time  Term_pre Rating     Cumprob
1   pre spring_24      1 0.017603242
2  post spring_24      1 0.007381687
3   pre spring_25      1 0.012442577
4  post spring_25      1 0.008026454
5   pre spring_24      2 0.085884845
6  post spring_24      2 0.037529326
7   pre spring_25      2 0.061969075
8  post spring_25      2 0.040699418
9   pre spring_24      3 0.299446988
```

```
10 post spring_24      3 0.150668895
11  pre spring_25      3 0.231096356
12 post spring_25      3 0.161789367
13  pre spring_24      4 0.879245310
14 post spring_24      4 0.751358540
15  pre spring_25      4 0.836594485
16 post spring_25      4 0.766787795
17  pre spring_24      5 1.000000000
18 post spring_24      5 1.000000000
19  pre spring_25      5 1.000000000
20 post spring_25      5 1.000000000

  time  Term_pre Rating
1  pre spring_24      4
2 post spring_24      4
3  pre spring_25      4
4 post spring_25      4

$Models

Model: "clmm, Value ~ time * Term_pre + (1 | Participant_ID), s_24_25_MIGS_072525_peer_longer
Null:  "clmm, Value ~ 1 + (1 | Participant_ID), s_24_25_MIGS_072525_peer_longer"

$Pseudo.R.squared.for.model.vs.null
                             Pseudo.R.squared
McFadden                            0.0158261
Cox and Snell (ML)                  0.0394710
Nagelkerke (Cragg and Uhler)        0.0428336

$Likelihood.ratio.test
 Df.diff LogLik.diff  Chisq    p.value
      -3     -13.169 26.337 8.1058e-06

$Number.of.observations

Model: 654
Null:  654

$Messages
[1] "Note: For models fit with REML, these statistics are based on refitting with ML"

$Warnings
```

```
[1] "None"
```

Significant effect for time.

Checking for Assumptions & Model Diagnostics

Brand Test: Proportional Odds Assumption

```
------------------------------------------------------------
Test for            X2  df  probability
------------------------------------------------------------
Omnibus             10.87   9   0.28
timepost            8.93    3   0.03
Term_prespring_25       1.43    3   0.7
timepost:Term_prespring_25  2.3 3   0.51
------------------------------------------------------------

H0: Parallel Regression Assumption holds
```

Time post pretty low

Informal checks with GLMM– not sure that doing correctly

Convergence issue with first model

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value_dummy <= -2) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_peer_longer_4dummy

      AIC       BIC    logLik -2*log(L)  df.resid
    127.7     150.2     -58.9     117.7       649

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.44086 -0.00958 -0.00766 -0.00766  1.94483

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 86.5     9.3
Number of obs: 654, groups:  Participant_ID, 327
```

```
Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -9.2822     1.2523  -7.412 1.24e-13 ***
timepost                    -0.4487     0.9559  -0.469    0.639
Term_prespring_25           -0.0197     1.8929  -0.010    0.992
timepost:Term_prespring_25  -1.6121     2.4457  -0.659    0.510
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.254
Trm_prsp_25 -0.327  0.201
tmpst:T__25  0.188 -0.382 -0.239


Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value_dummy <= -1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_peer_longer_4dummy

      AIC       BIC    logLik -2*log(L)  df.resid
    396.7     419.1    -193.4     386.7       649

Scaled residuals:
     Min       1Q   Median       3Q      Max
-1.24906 -0.03281 -0.02135 -0.02135  1.76048

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 48.12    6.937
Number of obs: 654, groups:  Participant_ID, 327

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -6.7605     0.7489  -9.027   <2e-16 ***
timepost                    -0.8596     0.5163  -1.665   0.0959 .
Term_prespring_25            0.1483     0.8455   0.175   0.8607
timepost:Term_prespring_25  -0.7163     1.0801  -0.663   0.5072
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.062
Trm_prsp_25 -0.299  0.190
tmpst:T__25  0.170 -0.446 -0.303

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: I(value_dummy <= 0) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_peer_longer_4dummy

      AIC       BIC    logLik -2*log(L)  df.resid
    737.5     759.9    -363.7     727.5       649

Scaled residuals:
    Min      1Q  Median      3Q     Max
-0.9243 -0.5369 -0.3417  0.6665  2.0240

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 1.69     1.3
Number of obs: 654, groups:  Participant_ID, 327

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -0.7373     0.1900  -3.880 0.000104 ***
timepost                    -1.2528     0.2591  -4.835 1.33e-06 ***
Term_prespring_25           -0.3670     0.3627  -1.012 0.311581
timepost:Term_prespring_25   0.6765     0.4770   1.418 0.156059
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.343
Trm_prsp_25 -0.422  0.282
tmpst:T__25  0.219 -0.510 -0.586

Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
```

```
 Family: binomial  ( logit )
Formula: I(value_dummy <= 1) ~ time * Term_pre + (1 | Participant_ID)
   Data: s_24_25_MIGS_072525_peer_longer_4dummy

      AIC       BIC    logLik -2*log(L)  df.resid
    722.4     744.8    -356.2     712.4       649

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.9098  0.3209  0.4061  0.4667  0.7614

Random effects:
 Groups         Name        Variance Std.Dev.
 Participant_ID (Intercept) 1.419    1.191
Number of obs: 654, groups:  Participant_ID, 327

Fixed effects:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                  1.8868     0.2489   7.580 3.47e-14 ***
timepost                    -0.7487     0.2441  -3.067  0.00216 **
Term_prespring_25           -0.5150     0.3741  -1.377  0.16863
timepost:Term_prespring_25   0.5914     0.4646   1.273  0.20305
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) timpst Tr__25
timepost    -0.661
Trm_prsp_25 -0.475  0.374
tmpst:T__25  0.332 -0.520 -0.657


        (Intercept)   timepost Term_prespring_25 timepost:Term_prespring_25
P_25_1   -9.2910611 -0.4487234       -0.01970051                 -1.6120964
P_25_11  -6.8329807 -0.8596354        0.14834521                 -0.7163127
P_25_12  -0.1224094 -1.2527697       -0.36704194                  0.6765437
P_25_13   2.3171903 -0.7486668       -0.51496915                  0.5914484


                (Intercept)                   timepost
                  3.4823152                  0.8274488
          Term_prespring_25 timepost:Term_prespring_25
                  0.1883416                  0.2651042
```

Checking other assumptions

```
formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_peer_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  654  -834.79 1683.58 5(0)  6.17e-08 4.4e+01

Coefficients:
                  timepost          Term_prespring_25
                    0.7315                     0.2848
timepost:Term_prespring_25
                   -0.3872

Threshold coefficients:
 -2|-1   -1|0    0|1    1|2
-3.288 -1.831 -0.614  1.532
```

```
formula: Value ~ time * Term_pre
data:    s_24_25_MIGS_072525_peer_longer

 link  threshold nobs logLik  AIC     niter max.grad cond.H
 logit flexible  654  -834.79 1683.58 5(0)  6.17e-08 4.4e+01

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
timepost                     0.7315     0.1703   4.295 1.75e-05 ***
Term_prespring_25            0.2848     0.2410   1.182    0.237
timepost:Term_prespring_25  -0.3872     0.3394  -1.141    0.254
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Threshold coefficients:
      Estimate Std. Error z value
-2|-1  -3.2884     0.2585 -12.722
-1|0   -1.8312     0.1524 -12.013
0|1    -0.6140     0.1235  -4.972
1|2     1.5317     0.1383  11.077
```

Tests of scale effects

```
formula: Value ~ time * Term_pre
```

```
              Df  logLik    AIC     LRT Pr(>Chi)
<none>           -834.79 1683.6
time           1 -834.75 1685.5 0.09149   0.7623
Term_pre       1 -834.78 1685.6 0.01726   0.8955
time:Term_pre  3 -834.43 1688.9 0.72342   0.8677

Tests of nominal effects

formula: Value ~ time * Term_pre
              Df  logLik    AIC    LRT Pr(>Chi)
<none>           -834.79 1683.6
time           3 -831.59 1683.2 6.4112  0.09323 .
Term_pre       3 -833.63 1687.3 2.3165  0.50937
time:Term_pre
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Likelihood ratio tests of cumulative link models:

         formula:                                        link: threshold:
peer_fm  Value ~ time * Term_pre                         logit flexible
Peer_fmm Value ~ time * Term_pre + (1 | Participant_ID) logit flexible

         no.par    AIC  logLik LR.stat df Pr(>Chisq)
peer_fm       7 1683.6 -834.79
Peer_fmm      8 1653.8 -818.91  31.756  1  1.748e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

[1] 1653.828

[1] 1780.94


spring_24


Descriptive Statistics  (N=492)

+----------+-----------+-----------+
|          |pre        |post       |
```

```
|          |(N=246)    |(N=246)    |
+----------+-----------+-----------+
|Value : -2|0.03  (  8)|0.03  (  7)|
+----------+-----------+-----------+
|    -1    |0.09  ( 21)|0.06  ( 15)|
+----------+-----------+-----------+
|    0     |0.25  ( 62)|0.09  ( 23)|
+----------+-----------+-----------+
|    1     |0.44  (109)|0.52  (128)|
+----------+-----------+-----------+
|    2     |0.19  ( 46)|0.30  ( 73)|
+----------+-----------+-----------+
```

spring_25

Descriptive Statistics (N=162)

```
+----------+----------+----------+
|          |pre       |post      |
|          |(N=81)    |(N=81)    |
+----------+----------+----------+
|Value : -2|0.02  ( 2)|0.00  ( 0)|
+----------+----------+----------+
|    -1    |0.10  ( 8)|0.07  ( 6)|
+----------+----------+----------+
|    0     |0.19  (15)|0.15  (12)|
+----------+----------+----------+
|    1     |0.44  (36)|0.49  (40)|
+----------+----------+----------+
|    2     |0.25  (20)|0.28  (23)|
+----------+----------+----------+
```

Check for Multicollinearity

Low Correlation

```
          Term  VIF   VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
          time 2.71 [2.40, 3.08]     1.65      0.37     [0.33, 0.42]
      Term_pre 3.09 [2.73, 3.52]     1.76      0.32     [0.28, 0.37]
 time:Term_pre 2.85 [2.52, 3.24]     1.69      0.35     [0.31, 0.40]
```